

MAST-U | NBI Power Scaling

Neutral Beam Injection & Neutron Response Analysis

8

Shots Analysed

9

NBI Segments

3.24

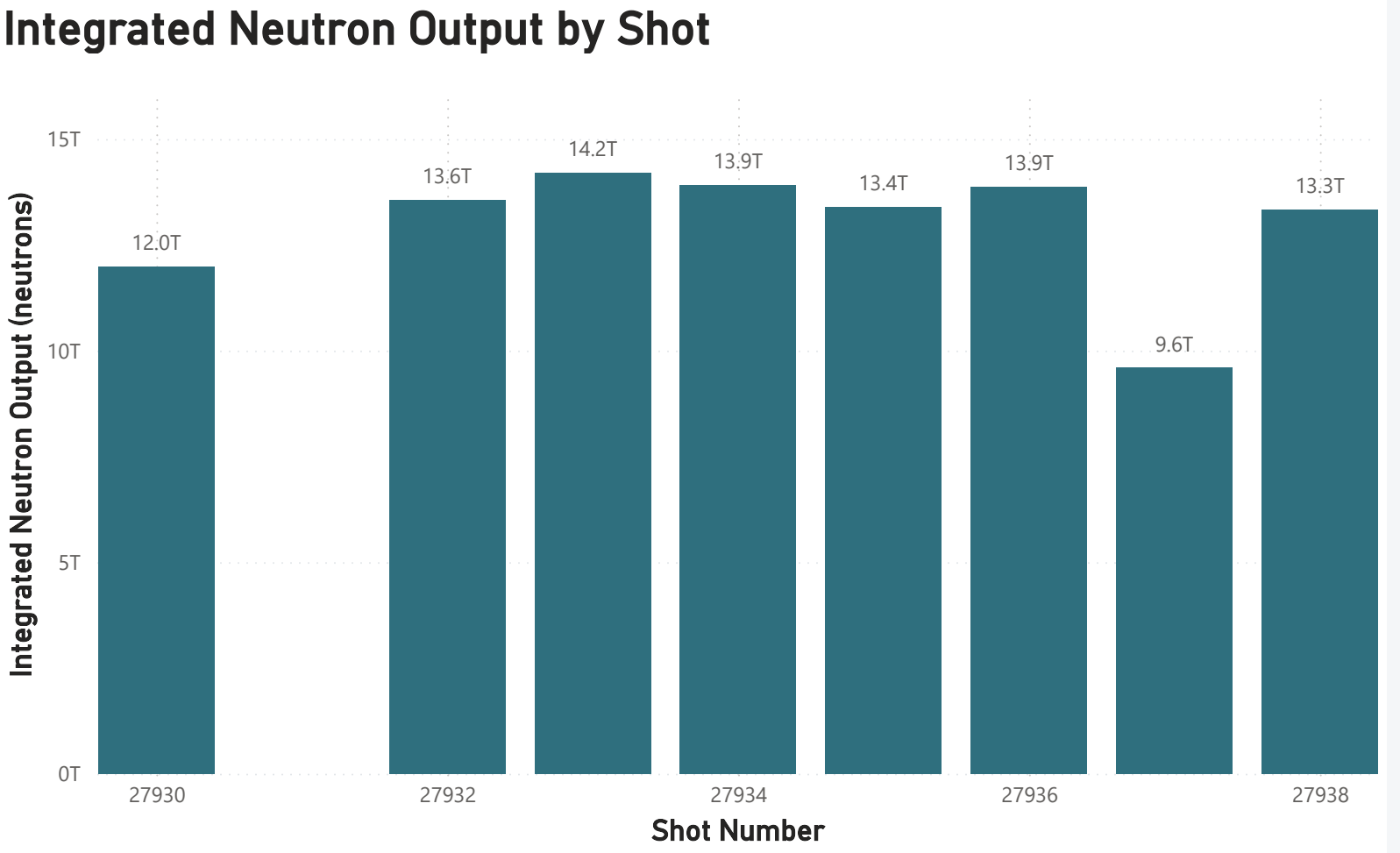
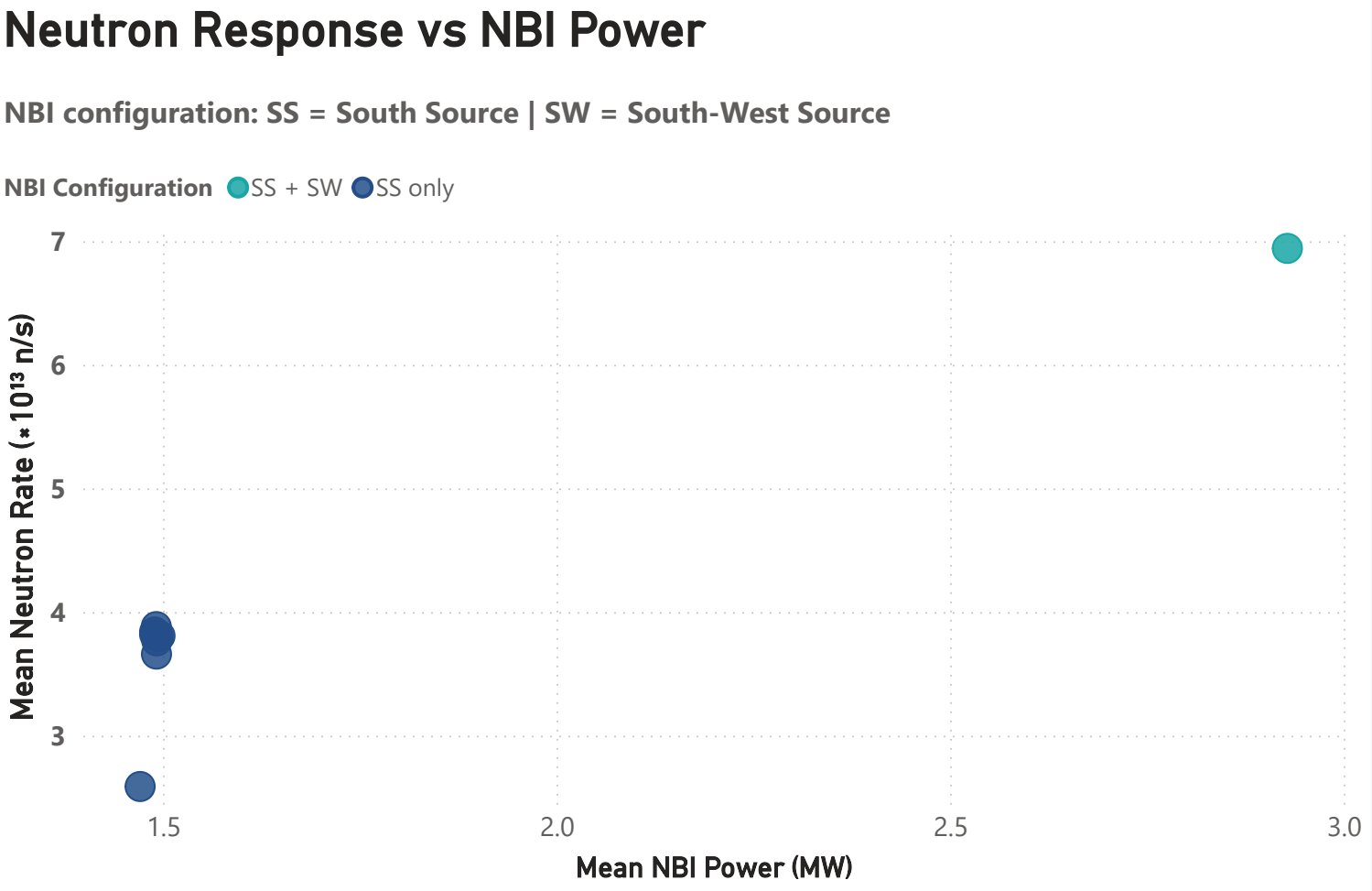
Peak NBI Power (MW)

92.90T

Peak Neutron Rate (n/s)

Select Shot

All



NBI Configuration

8

SS Only

1

SS + SW

Neutron Output Range

9.59T

Minimum Neutron Output

14.20T

Maximum Neutron Output

NBI Power Range (MW)

1.47

Minimum NBI Power

2.93

Maximum NBI Power

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NBI Segment & Plasma Response Analysis

9

Segments Selected

1.65

Mean NBI Power (MW)

40.11T

Mean Neutron Rate (n/s)

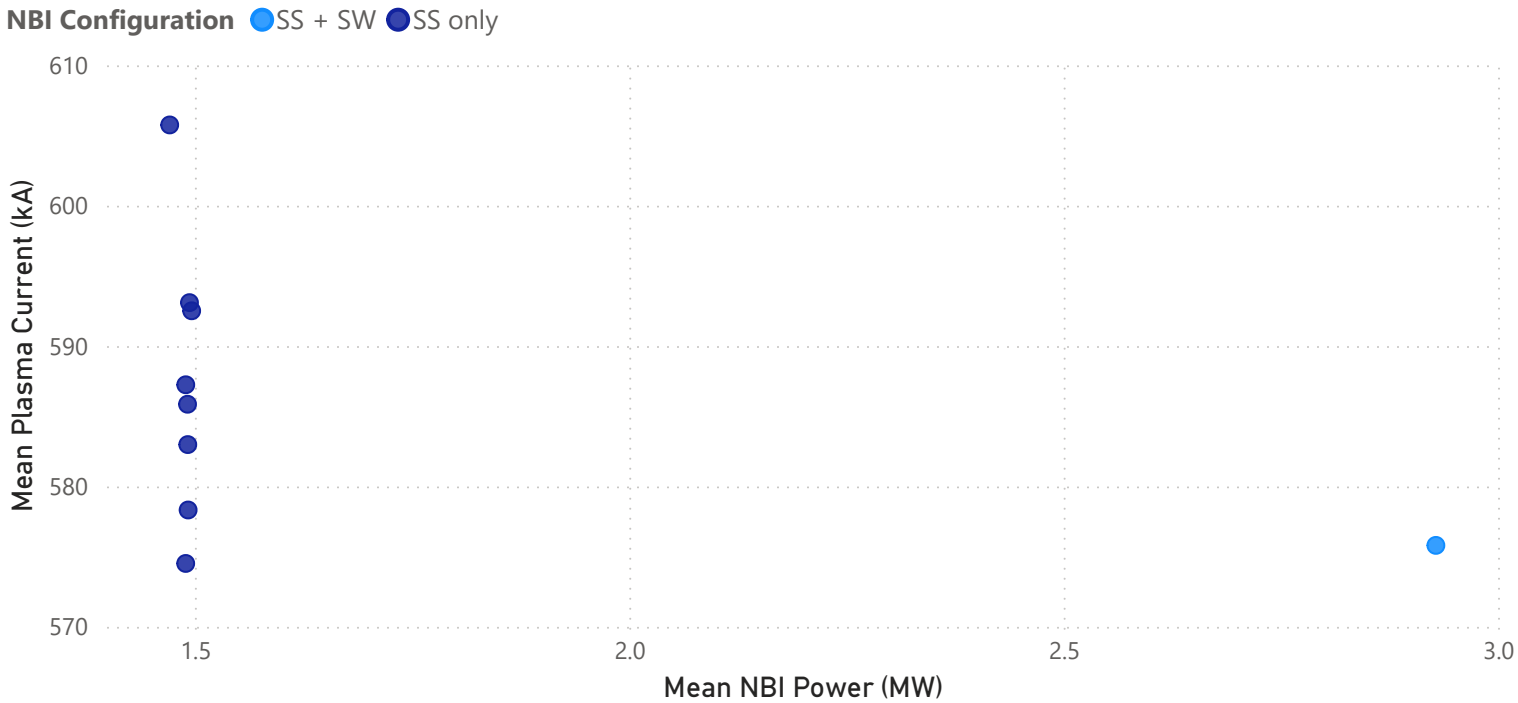
586.23

Mean Plasma Current (kA)

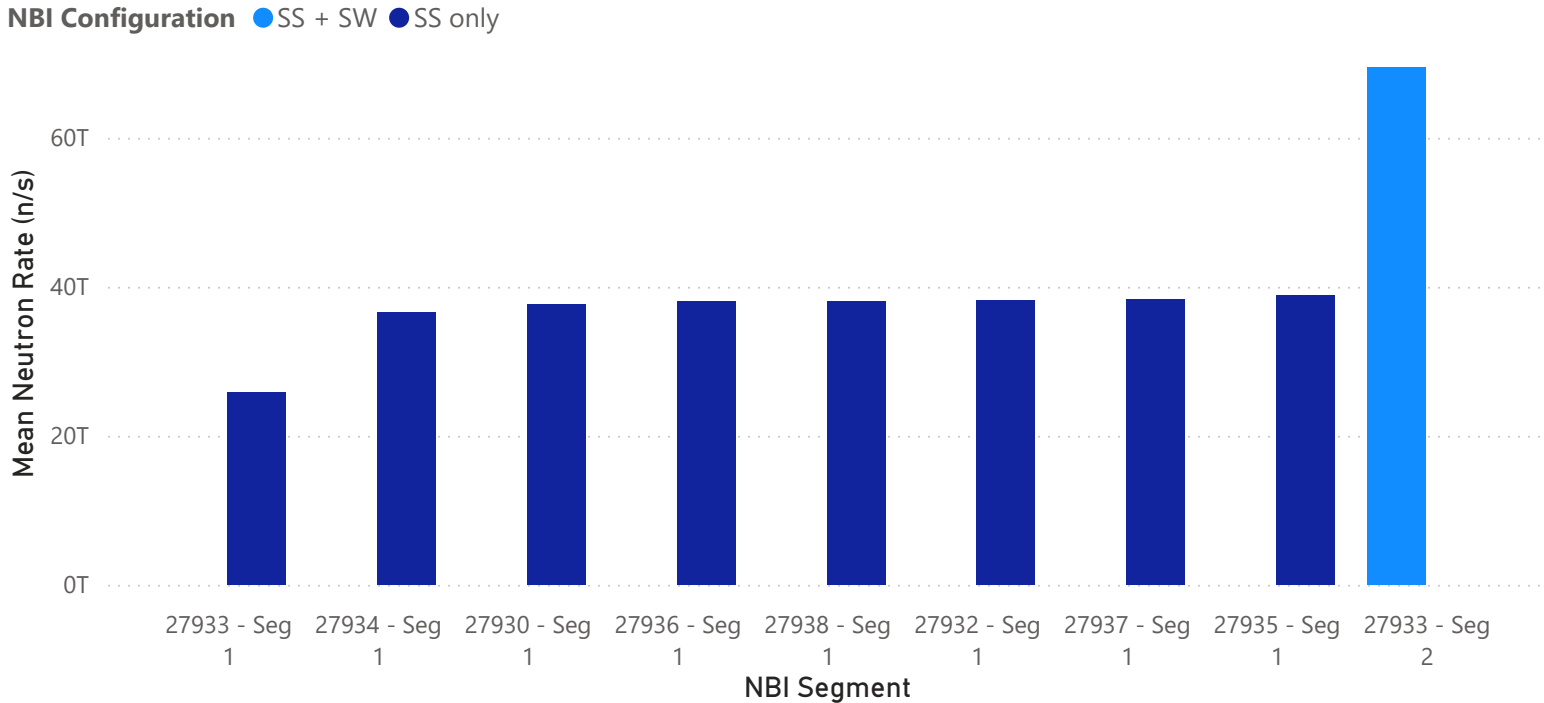
Select Shot

All

Plasma Current vs NBI Power



Mean Neutron Rate by NBI Segment



NBI Segment Details

Shot	Segment	NBI Configuration	Start (s)	End (s)	Duration (s)	Mean NBI Power (MW)	Mean Neutron Rate (n/s)	Mean Plasma Current (kA)
27933	1	SS only	0.15	0.20	0.05	1.47	25.88T	605.75
27937	1	SS only	0.20	0.46	0.26	1.49	38.38T	574.51
27932	1	SS only	0.15	0.52	0.37	1.49	38.18T	587.24
27935	1	SS only	0.15	0.51	0.36	1.49	38.82T	585.85
27934	1	SS only	0.15	0.54	0.39	1.49	36.59T	582.98
27930	1	SS only	0.17	0.50	0.33	1.49	37.66T	578.32
27936	1	SS only	0.15	0.53	0.38	1.49	38.00T	593.09
27938	1	SS only	0.15	0.52	0.37	1.50	38.07T	592.51
27933	2	SS + SW	0.20	0.40	0.20	2.93	69.42T	575.80

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Individual Shot & NBI Segment Analysis

Select Shot

All

▼

NBI Segments

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Peak NBI Power (MW)

2.93

Peak Neutron Rate

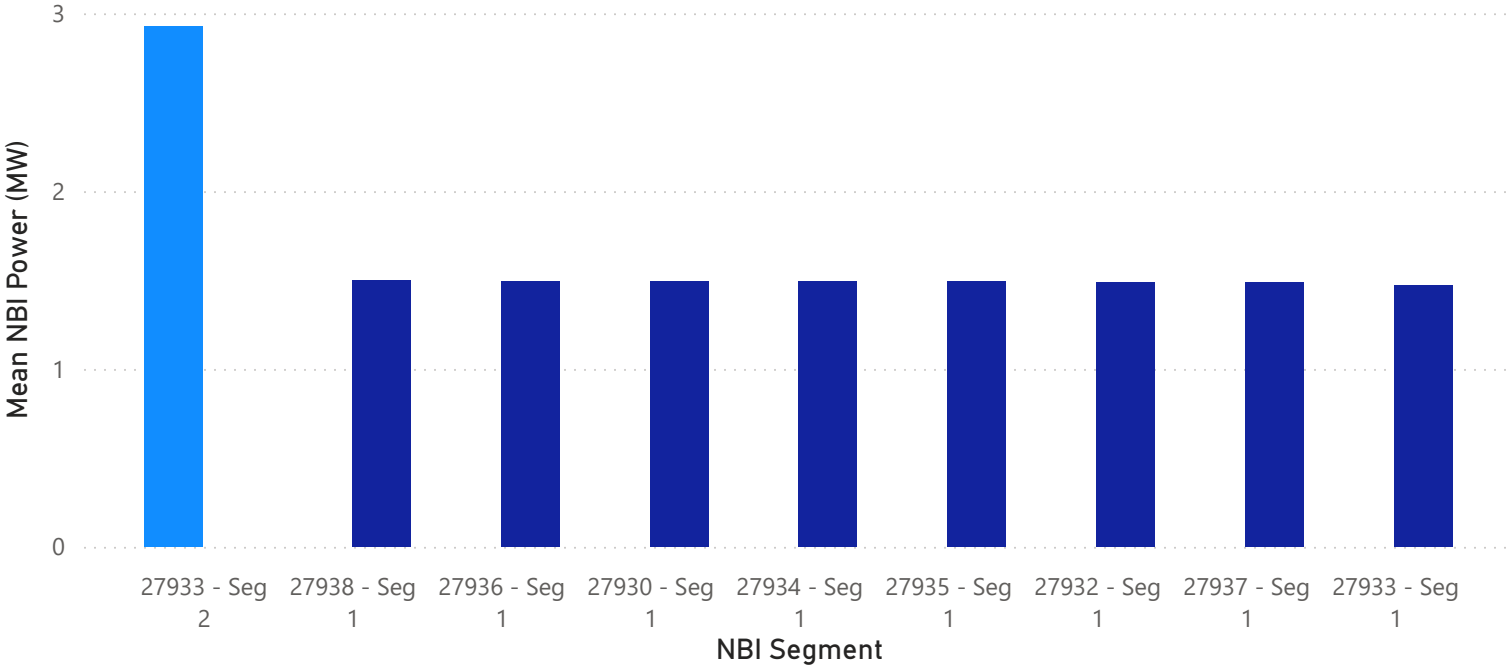
69.42T

Mean Plasma Current (kA)

586.23

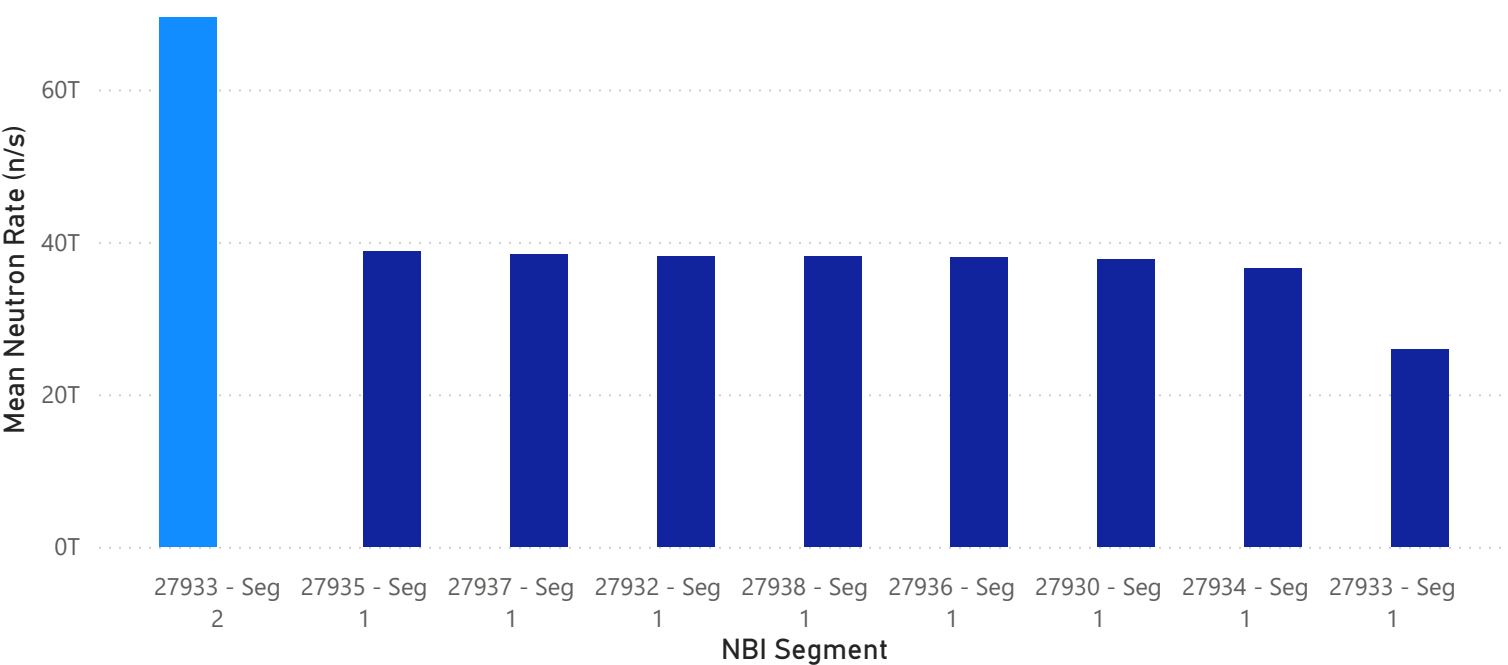
NBI Power by Segment

NBI Configuration ● SS + SW ● SS only



Neutron Response by Segment

NBI Configuration ● SS + SW ● SS only



Segment	NBI Configuration	Start (s)	End (s)	Duration (s)	Mean NBI Power (MW)	Mean Neutron Rate (n/s)	Mean Plasma Current (kA)
2	SS + SW	0.20	0.40	0.20	2.93	69.42T	575.80
1	SS only	0.15	0.52	0.37	1.49	38.18T	587.24
1	SS only	0.15	0.20	0.05	1.47	25.88T	605.75
1	SS only	0.15	0.51	0.36	1.49	38.82T	585.85
1	SS only	0.15	0.52	0.37	1.50	38.07T	592.51
1	SS only	0.15	0.53	0.38	1.49	38.00T	593.09
1	SS only	0.15	0.54	0.39	1.49	36.59T	582.98
1	SS only	0.17	0.50	0.33	1.49	37.66T	578.32
1	SS only	0.20	0.46	0.26	1.49	38.38T	574.51

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Signal Variability & Data Quality Assessment

Segments Analysed

9

Mean NBI Power SD (MW)

0.09

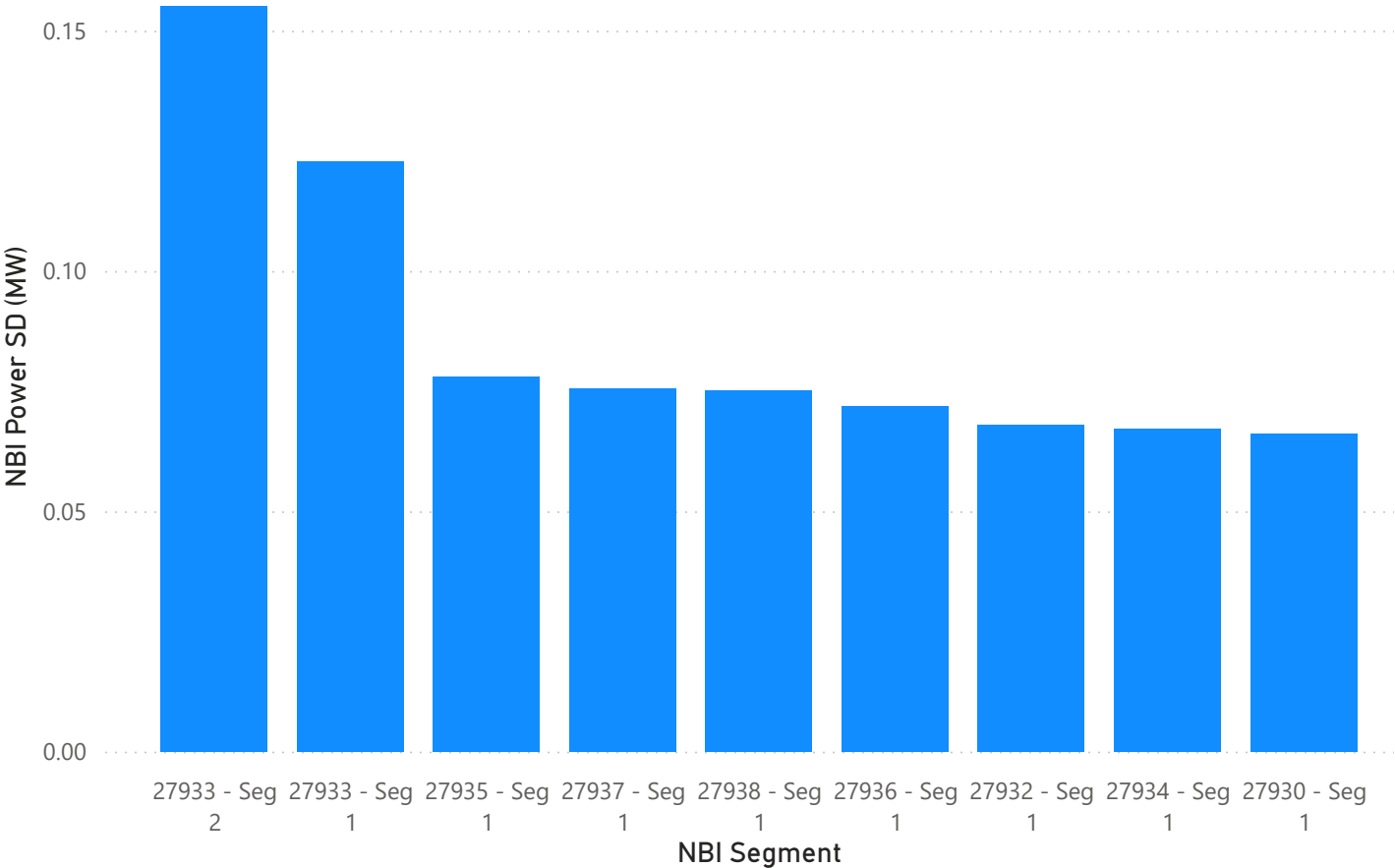
Mean Neutron Rate SD (n/s)

52.26T

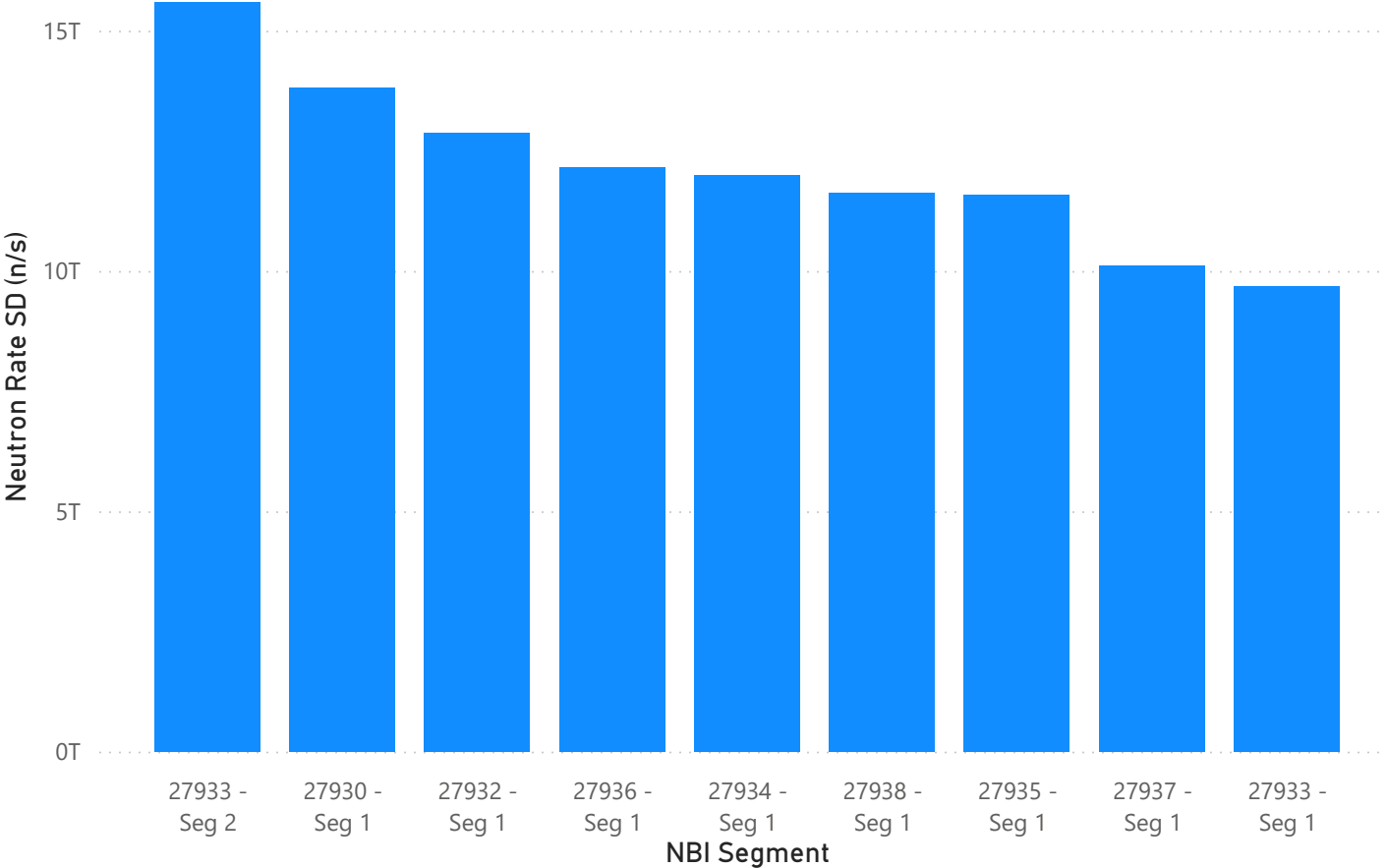
Mean Plasma Current SD (kA)

108.75

NBI Power Variability by Segment



Neutron Rate Variability by Segment



NBI Power CV (%)

5.25

Neutron Rate CV (%)

31.32

Plasma Current CV (%)

18.64